

Test Strategy — peviitor.ro

Project: peviitor.ro — open source job search engine for Romania

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Document Approval

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1. SCOPE

1.1 Purpose of the Document

This document defines the test strategy for the **peviitor.ro** platform — the open source job search engine for Romania. A core strategic principle is that the platform is built exclusively with **open source and free technologies**, and this commitment is not subject to change. Its purpose is to establish a unified testing approach covering all major platform components, from front-end to the indexing infrastructure, with the following scope objectives for each area:

Apache SOLR

- Upgrade to the latest stable version
- Maintain two dedicated cores: `job` and `company`
- Ensure the SOLR schema matches the Job and Company models defined in `peviitor_core`
- Guarantee data consistency — no orphaned fields, no missing mandatory fields from the job or company model
- Implement Basic Authentication as a security layer

PHP API (`api.peviitor.ro`)

- Maintain two versions: **v0** for development and testing, **v1** for production
- Keep the codebase simple, fast, and easy to maintain
- Ensure the code is easily deployable
- Ensure the code is testable
- Eliminate security vulnerabilities
- Secure all API endpoints

UI (`search-engine`)

- Keep the UI simple to write and maintain
- Ensure the UI conforms to the UX Design (Figma)
- Host the UI on GitHub Pages
- Ensure the UI remains accessible even if the BFF PHP API is down (graceful degradation)

Search Functionality

- Make the search interface easy to use
- Deliver relevant search results
- Provide useful filters for users (location, company, tags, work mode)

1.2 Out of Scope

- Individual scraper testing (each scraper is handled separately by the scraping teams)
- Validator (admin.peviitor.ro) — will be covered in a separate test strategy document
- Mobile application testing (androidAPP) — will be covered in a separate test strategy for mobile testing

1.3 General Objectives

- Ensure the quality of aggregated data (jobs, companies)
- Validate search engine functionality (SOLR)
- Confirm the correctness of the REST API (api.peviitor.ro)
- Verify the user experience in the web interface (search-engine)
- Identify defects early through manual and automated testing
- Ensure availability and performance on the existing infrastructure (Raspberry Pi 5, CloudFlare)
- Handle errors gracefully — clear messages for no results, invalid input, unavailable external links
- Validate input handling — long input (>500 chars), special characters, empty search, case sensitivity
- Verify correct integration with external links (LinkedIn, GitHub, Discord, etc.)

1.4 Quality Benefits Targeted

- **Functionality:** All features work as specified
- **Performance:** Response times < 2s for searches, daily indexing
- **Security:** OWASP Top 10 compliance, input sanitization, API endpoint security
- **Reliability:** Correct data, no dead links, expired jobs automatically removed
- **Resilience:** Platform remains operational under stress (API down, SOLR offline)
- **Disaster Recovery:** SOLR index backup and restoration
- **Redundancy:** Failover mechanisms for critical components
- **Scalability:** Ability to aggregate 40,000+ jobs updated daily
- **Validity:** Compliance with the data schema (Job Model, Company Model)
- **Accessibility:** WCAG 2.2 Level AA compliance
- **Usability:** Intuitive search flow, clear filters, helpful error messages, mobile-friendly experience
- **Browser Compatibility:** Latest 2 versions of Chrome, Firefox, Edge, Safari
- **Mobile Responsiveness:** All pages functional on viewports ≥ 320px
- **SEO:** Valid meta tags, semantic HTML, proper heading hierarchy (h1-h6)